Root Cause Analysis Report

Part	MS-4505-001 Sheet Metal Bracket
Material / process	SPCC Steel / bending
Case	rca_512bdc6b288e450b8b830d6388d4f4af
Date	2026-11-01 17:57:18
Status	AI proposal, pending expert review

1. Problem

Defect	Severity	Confidence	Detail
scratch	minor	95%	46.1 mm long
dent	major	88%	~14 mm
deformation	major	99%	bend 93.1 deg (nom 90 deg ±0.5)

2. Root cause

Cause: Tool Wear / Die Damage

Reason: Detected defects include a minor scratch and a major dent on the SPCC steel bracket. Evidence [587aa665-c6f1-48df-8783-09d65ff4f82a] confirms that progressive wear or damage on punch and die tooling directly leads to surface defects such as scratches from worn edges and dents from damaged die surfaces during bending operations.

Effect: Surface defects (scratches from worn tooling edges), dents from damaged die surfaces

Confidence: 95%

Supporting evidence
- **587aa665-c6f1-48df-8783-09d65ff4f82a**: Tool Wear / Die Damage Root Cause Analysis Defect Types: scratch, dent, edge burr, dimensional shift Process: bending, stamping, forming Root Cause: Progressive wear of punch/die tooling causes dimensional drift, surface scratches, edge burrs and dents on formed parts. Effect: Surface defects (scratches from worn tooling edges), dents from damaged die surfaces, edge burrs from worn cutting edges,

3. Solution

Corrective actions (fix this occurrence)

1. Inspect tooling under magnification for wear, chipping, or damage
2. Regrind worn punch/die to restore geometry within tolerance

Preventive actions (stop it recurring)

1. Implement routine tool inspection and maintenance schedules based on stroke counts
2. Establish clear wear limits for punch and die components

Precautions

- Inspect tooling under magnification for wear, chipping, or damage
- Regrind worn punch/die to restore geometry within tolerance
- Implement routine tool inspection and maintenance schedules based on stroke counts
- Establish clear wear limits for punch and die components

4. Other causes considered

Rank	Cause	Confidence	Reason

2	Improper Bending Force / Over Bending	95%	A major deformation defect was detected on the part. Evidence [586f086d-0deb-40c2-a71d-f840aaa55a60] shows that applying improper bending force or over-bending beyond the material elastic limit causes deformation of sheet metal components during press brake operations.
3	Springback	54%	Major deformation was observed on the sheet metal bracket. Evidence [6872e1ff-9d0f-41f8-9cd9-c783d66c1b46] explains that elastic recovery after sheet metal forming (springback) causes deformation and dimensional deviations, though specific angular out-of-tolerance features were not noted in the dete

5. Traceability

AI model: gemini:gemini-3.5-flash. Expert edited the cause: no.
This analysis was drafted with AI assistance from the retrieved knowledge-base evidence. It must be reviewed and confirmed by a qualified quality expert before corrective action is taken.